From activity to evidence

What ran?

What human work happened?

What can we responsibly predict?

OBSERVE

MEASURE

CHECK

An engineer's guide to the model, its implementation and its limits

A tool ran for 20 minutes. How much human work?

20 min is a tool-runtime observation.

Unattended

The engineer does other work while the tool runs.

Supervised

The engineer watches, diagnoses or steers the run.

Collaborative

Several people review or debug during the same interval.

A runtime receipt cannot choose between these human-work histories.

Three questions need three kinds of evidence

What executed?

Receipts + GitHub metadata

Tool identity, outcome and reported resources

What work and output?

Human observations + assessments

Person-hours and retained requirement credit

What might merge?

Historical PR counts

An empirical forecast distribution

Connect the records; preserve the meaning of each quantity.

Follow one requirement through many activities

Requirement R-17: export a report with verified checksums.



Agree the acceptance rule	Human + assistant work	Tests, failures and retries	Record explicit acceptance	Assess retained function

One requirement can have many executions—and unsuccessful work still counts.

Keep the source, the interpretation and the cut

1 Source bytes

What was retrieved or imported?

Retain the original bytes.

2 Validated records

What does this record assert?

Keep provenance and identity.

3 Frozen snapshot

What was known at this cut?

Preserve reproducible results.

Later evidence creates a new view; it does not silently rewrite an old one.

The GitHub account is not the controller

Platform identity

Original actor

Rerun actor

Trigger event

Declared controller

human · assistant

automation · unknown

Supporting provenance

Session linkage

Declared study session

Logical call and parent

Tool-reported session, separately

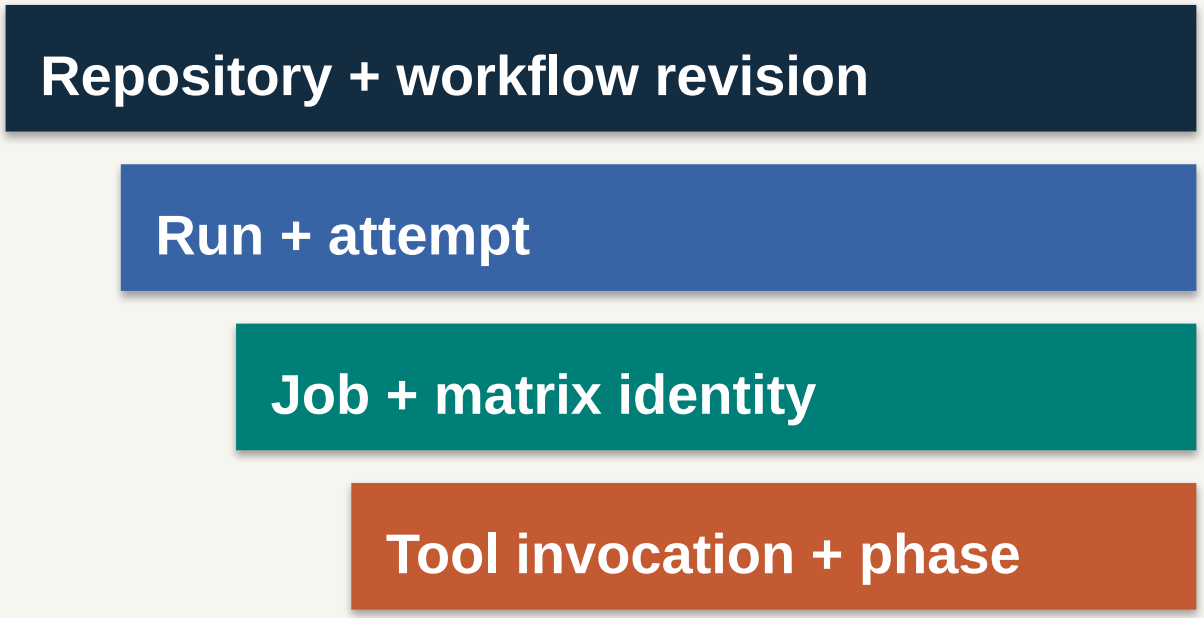
Keep unknown when provenance is unavailable; do not infer it from the account.

The same account can hide two different stories

Scenario	GitHub actor	Declared controller	Tool
Assistant uses Alice's credentials	Alice	assistant	cargo test
Alice launches an AI workflow	Alice	human	Codex

Credentials, control and tool type are separate observations.

Identify an execution at the right level



Where and which definition?

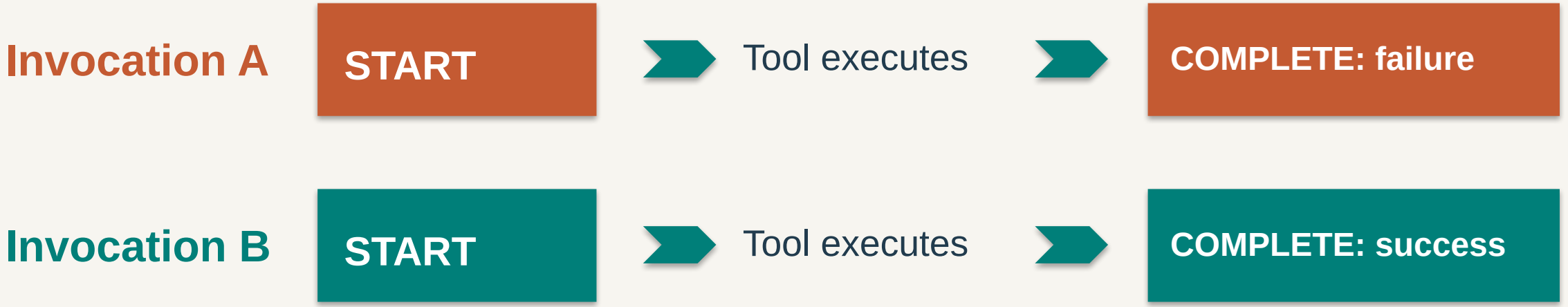
Original execution or rerun?

Which parallel worker?

Which execution, start or completion?

Repeated delivery reuses identity. A real retry gets a new invocation ID.

Write a start before the tool can fail



Same logical call; distinct actual invocations.

Never overwrite the failed attempt with the successful retry.

A wrapper observes only the boundary you wrap

One setup step cannot automatically see every tool or every job.

EXPLICITLY WRAPPED PROCESS

`cargo test`

Start · exit status

Runtime · named check

Still outside this receipt:

Internal tool operations

Other jobs

Work outside Actions

Describe the boundary before interpreting the evidence.

Three components cooperate

1 Job recorders

Explicit wrappers

Separate per-job bundles

Best-effort publication

2 Durable collector

Download + validate

Deduplicate + retain

One durable evidence store

3 Reconciliation

Independent GitHub metadata

Compare runs and jobs

Keep observation gaps

Runner-local recording and durable retention solve different problems.

A missing receipt is an observation gap

What arrived?	What the model can say
Start + completion	A recorded invocation has both phases.
Start only / completion only	Retain the phase; flag the missing observation.
Valid empty bundle	The bundle reports setup context with zero receipt entries.
No bundle	No receipts observed; tool use remains unknown.

Keep failed, cancelled and skipped GitHub jobs—even when no bundle arrives.

A receipt answers six practical questions

Question	Evidence retained
Which execution?	Repository, workflow revision, run/attempt, job/matrix, invocation
Who or what initiated it?	Actors + event; separate controller and provenance declarations
Which tool and purpose?	Tool/version, configuration, development or product execution
What happened?	Start/end, exit status, checks, runtime, reported usage and charges
What does it relate to?	Protocol/task, declared session, logical call, parent invocation
Which artifacts, and what is missing?	Missing fields and designated input/output artifact hashes

Configured, reported and unknown stay separate

Configured

The caller selected a model and supplied configuration metadata.

Tool-reported

The adapter recognised a thread ID or token-usage field.

Unknown

No recognised observation of model identity, charge or usage was available.

The initial Codex adapter leaves observed model identity unknown.

A fingerprint checks bytes. Evidence checks claims.

Hash + schema

Are these bytes unchanged?

Does the record satisfy the contract?

Independent comparison

Does the API support the platform context and job linkage?

Human or tool provenance

What supports the remaining declarations and observations?

Validation is a specific check—not a blanket certificate of truth.

Now link the execution to a requirement

Requirement R-17

Execution links

Tool receipts, PRs, revisions and sessions

Human work

Observed or sampled allocations for every included role

Outcome observations

Acceptance and retained functionality against the rubric

The execution layer supplies links; the measurement layer supplies its own observations.

Three clocks answer different questions

Elapsed acceptance

2 days

Calendar duration from enrolment to acceptance

Human labour

5 person-hours

Time allocated by people to the included work

Tool execution

20 minutes

Elapsed time inside the wrapped process

Always name the quantity and its unit before comparing numbers.

One person has one hour of capacity per hour

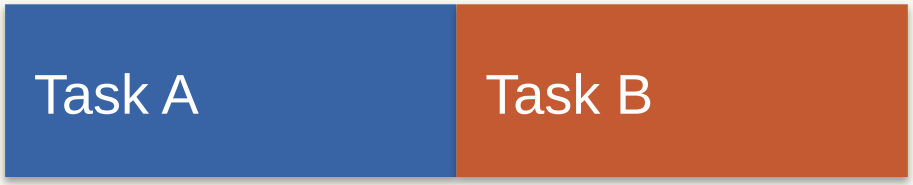
Two people pairing

1 hour allocated

1 hour allocated

1 + 1 = 2 person-hours

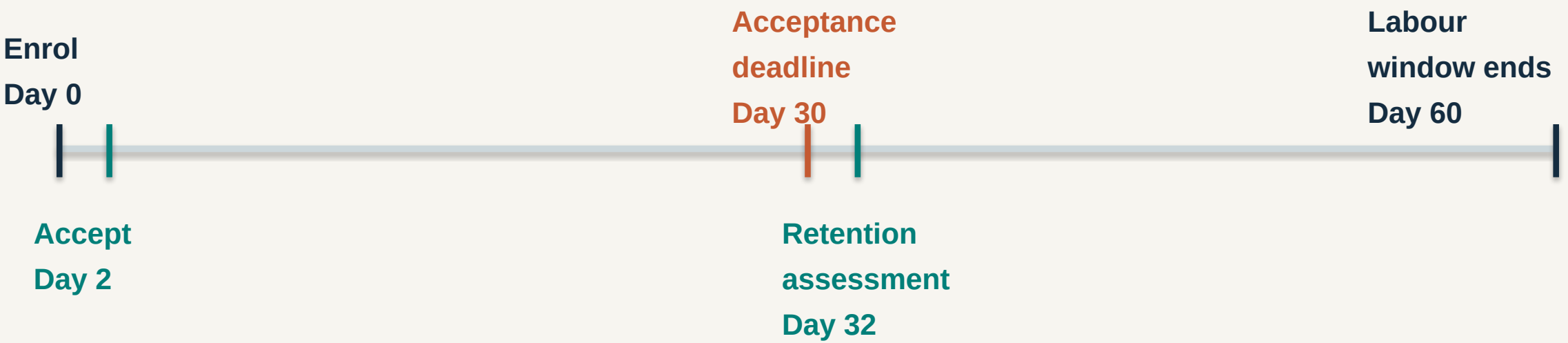
One person, two tasks



Combined allocation

≤ 1 person-hour

Freeze the deadlines before observing outcomes



Early acceptance does not shorten the cohort's labour window.

Count retained requirement credit, then divide

Task	Deadline outcome	Retained credit	Human labour
A	Accepted on time; 75% retained	0.75 units	2 h
B	Permanently abandoned	0	2 h
C	Accepted after the deadline	0	1 h

0.75 units ÷ 5 person-hours = 0.15 units/person-hour

> **Report both totals. Include the labour of unsuccessful work.**

Zero, unknown and pending are different states

Observed zero

Evidence and the frozen rule establish no counted contribution.

Unknown

The required observation or coverage is missing.

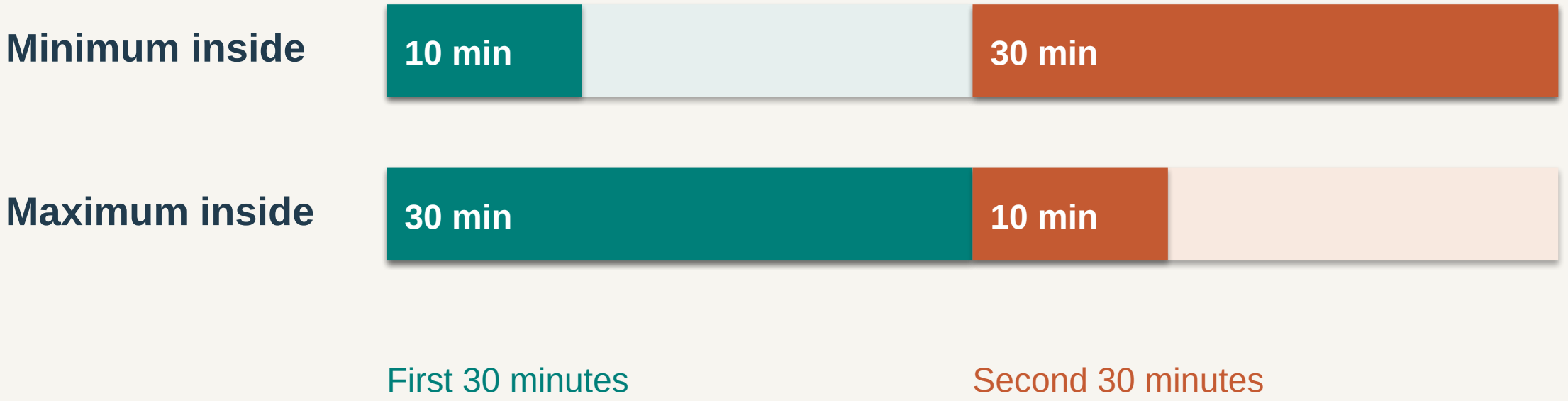
Pending

The observation window or follow-up has not yet closed.

Scope changes also affect protocol credit. Freeze that rule before intake and keep all applicable labour.

An interval total may leave its timing unknown

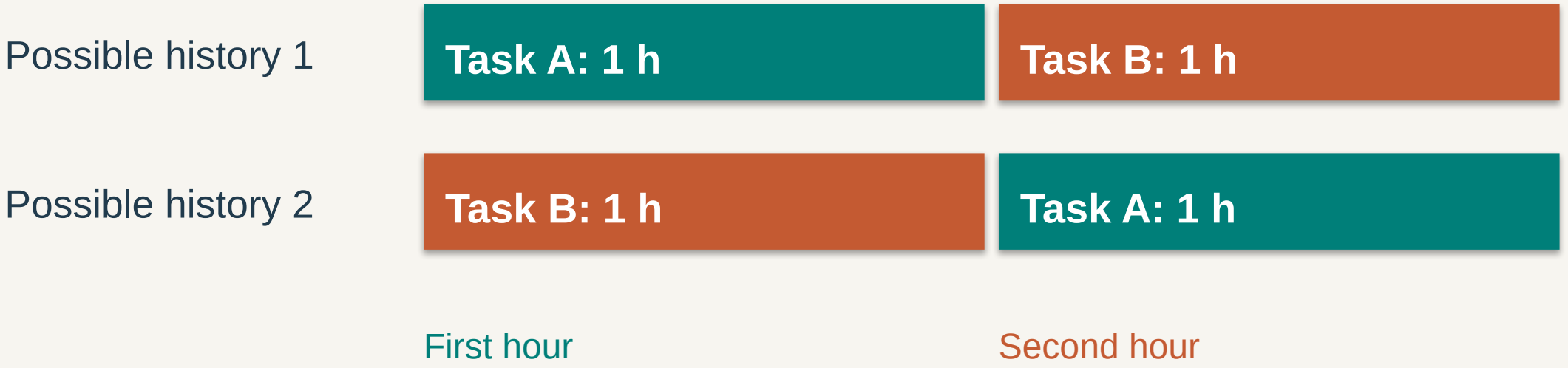
Known: 40 minutes of human work somewhere in one hour. Target: the first half-hour.



The justified answer is 10–30 minutes, not an invented exact timing.

Shared capacity can tighten the answer

One person. Two hours. Task A uses 1 h; task B uses 1 h.



Combined first-hour labour is exactly 1 h. The separate 0–2 h range is too wide.

Use a random sample to correct frozen predictions

Randomly select 2 of 4 intervals without replacement: each has a ½ chance.

Unit	Frozen prediction	Observed in sample	Error
1	10 min	14 min	+4 min
2	10 min	8 min	−2 min
3	10 min	Not selected	Unknown
4	10 min	Not selected	Unknown

40 + (2 × 4) + (2 × −2) = 44 minutes estimated

A bound and a sampling estimate answer different questions

Deterministic bounds

Which totals are possible under these observations and constraints?

Example: 10–30 min

Sampling estimate

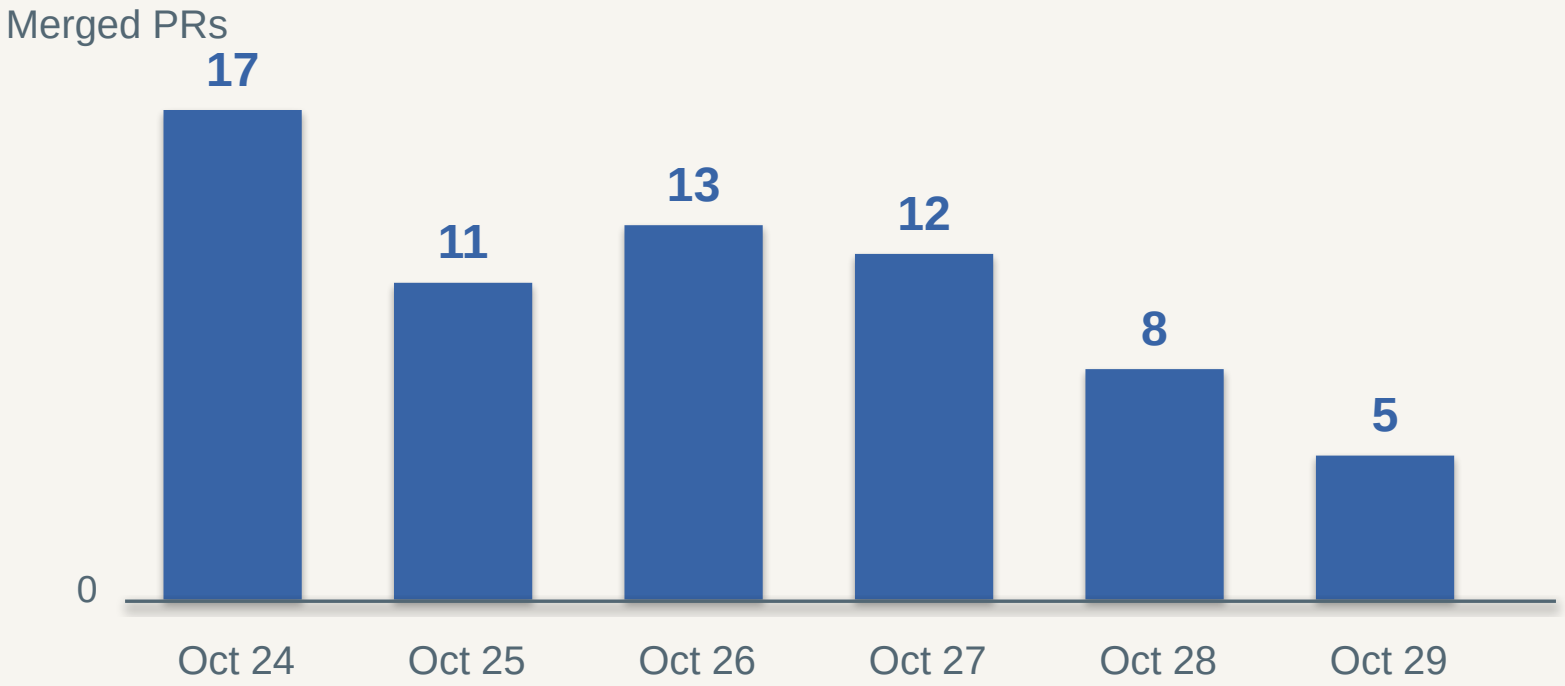
What does the sample estimate about the whole declared frame?

44 min; standard error ≈ 8.49 min

Uncertainty has a source. Name it, and retain the assumptions that support it.

A separate model forecasts merged-PR counts

Each complete historical day supplies one reference outcome.



11

PRs at the empirical median

Assumption: past days remain relevant.

This is a reference distribution—not a promise of eleven merges.

Preserve the awkward results

Daily range: 5–17

Contains 100% of the training mass.

Future coverage is unvalidated.

Earlier range hits: 0/2

Both reconstructed outcomes missed their ranges.

Keep this adverse evidence.

Weekly forecast

Unavailable: zero complete weeks.

Four complete weeks are required.

Do not multiply a daily prediction by seven to bypass missing weekly history.

More evidence does not automatically identify an AI effect

Observed

This tool completed.

This person-time was observed.

This requirement was assessed.

Descriptive

The cohort had this output and labour.

The historical counts had this distribution.

Causal

Would another policy have changed the outcome?

Requires a separate design.

Match the strength of the claim to the observations and study design.

The public pilot verified the evidence path

2

tool invocations

4

start/completion receipts

2

matched matrix jobs

Repeated collection: unchanged receipts. Retained store: 29 verified records.

The pilot supports instrumentation and replay—not a productivity claim.

Adopt one explicit boundary at a time

Instrument

Setup → explicit wrapper → publish

One bundle per job; known metadata only

Retain and reconcile

Collect before expiry

Validate, deduplicate and verify

Compare independent jobs

Measure deliberately

Freeze the study protocol

Observe people and outcomes

Link evidence to requirements

Start narrow, inspect the gaps, and expand only after the evidence survives replay.

Which claim can we defend?

Claim	What evidence would it need?
A The wrapped test completed successfully.	Completion receipt + supported job linkage
B This 20-minute run saved 20 minutes of human work.	Human observations + a justified comparison
C No tokens were reported, so no tokens were used.	An actual zero-usage observation
D Combined first-hour labour is exactly one hour.	The stated totals + shared capacity assumptions

Make the evidence strong enough for the claim

Collect what happened.

State how you know.

Keep the unknowns visible.

Evidence → explicit assumptions → a bounded, reproducible conclusion

Receipt fields and their provenance

Group	Key fields	Basis
platform	repository_id; workflow_sha; run_id; run_attempt; actors; event	Runner; selected fields corroborated
job context	job_key; job_name; job_id; matrix	Key: runner; name/ID/matrix: caller
declared	controller; provenance; protocol/task/session; logical/parent IDs	Caller declaration
tool	adapter; name/version; purpose; configured_model; configuration	Supplied labels/configuration
result	status; exit_code; runtime; observed model/session; tokens; charges	Wrapper / recognised tool report
artifacts / missing	role; label; SHA-256; size; missing_fields	Designated bytes / recorded absence

The implementation rejects ambiguous substitutions

Immutable identity

Same phase + same identity → same content

Retry → fresh invocation ID

Validated structure

Versioned schemas

Exact receipt hashes

Strict JSON and bounded ZIP

Preserved failures

Missing phase stays missing

Collection quarantines conflicts

Independent job evidence survives

16 MiB bundle • 10,000 receipts • 256 KiB per receipt • one bundle.json ZIP member

The small set of equations behind the pictures

Question	Rule	Read it as
Human contribution	duration × allocation	Time assigned to the included work
Timely credit, eligible scope	frozen score × retained fraction	Credit under the frozen rubric and scope rule
Cohort productivity	sum(output) ÷ sum(labour)	A ratio of totals; labour must be positive
Clipped interval	[max(0, m − v), min(m, u)]	Known mass m; u inside, v outside

For known Q ≥ 0 and 0 < Hmin ≤ H ≤ Hmax: the conditional ratio range is Q/Hmax to Q/Hmin.

Correct prediction errors with the actual sampling design

Estimate = prediction total + sum(error ÷ selection chance)

Required

Complete declared frame

Genuine selection probabilities

Accurate sampled labels

Predictions frozen in advance

What stays explicit

Nonresponse is rejected

Missing frame work is not repaired

Variance depends on the design

An estimate is not exact interval mass

The forecast is a reproducible reference, not a capacity model

Build the reference

Complete UTC bins

Four bins minimum

Portfolio counts summed first

No daily-to-weekly scaling

Score the original

Keep the issued distribution

Collect through the full target

Preserve losses and misses

Distinguish replay from issuance

Keep its scope

Merged PR count only

No labour or staffing inference

No causal queue diagnosis

Future skill remains unestablished

Sources for execution evidence

[X] Execution-evidence contract

[R] Receipt schema

[T] Public execution pilot and retained verification

[G] GitHub actor and triggering-actor contexts

[W] GitHub workflow_run completion event

[A] Architecture and evidence cuts

Content baseline: PR #9 implementation e252eae • 1 November 2026

Sources for measurement and forecasting

[P] Measurement protocol

[L] Exact labour identification

[J] Joint-capacity example input

[E] Model-assisted estimation

[F] Automatic forecasting contract

[D] Retained public forecast case, 30 October 2026

Content baseline: PR #9 implementation e252eae • 1 November 2026